

# Project Topic

**Title: Design and Implementation of a Personalized Machine Learning-Based Recommendation Engine.**

## **Abstract:**

With the explosive growth of digital platforms, users face severe choices overload, making it difficult to discover relevant content or products. This project proposes a standalone, intelligent Personalized Recommendation System designed to maximize user engagement and retention. The system utilizes machine learning algorithms, specifically combining Collaborative Filtering (analyzing user-behavior patterns) and Content-Based Filtering (analyzing item metadata and attributes). By transforming user profiles and item characteristics into high-dimensional vector embeddings, the system calculates cosine similarity to predict user preferences. The final deliverable is an optimized recommendation engine that takes user history as input and instantly outputs a highly accurate, personalized, and ranked list of recommended items, achieving high precision and recall metrics.